

Subject Description Form

Subject Code	ME5303
Subject Title	Robot Motion Planning
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil.
Objectives	1. To provide students with a comprehensive understanding of advanced spatio-temporal methods to plan the motion and trajectories of robotic systems. 2. To provide students with knowledge of sensor-based reactive algorithms that enable mobile robotic systems to navigate complex environments and avoid collisions with obstacles.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Mathematically formulate motion planning problems to represent practical robotic applications. b) Demonstrate proficiency in analysing and applying robot dynamic models to compute feasible trajectories. c) Design motion planning algorithms to generate collision-free paths for robotic systems in cluttered environments. d) Develop navigation strategies to enable autonomous robots to operate in dynamic environments. e) Apply numerically simulation tools to validate the performance of planning algorithms.
Subject Synopsis/ Indicative Syllabus	Principles of Motion Planning. Mathematical preliminaries, degrees-of-freedom, obstacles and constraints, configuration space, path planning algorithms, graph-based methods, sampling-based methods, model-based, gradient descent approaches, potential fields, optimal solutions for hyper-redundant manipulators and humanoid robots. Trajectory Planning. Kinematics and dynamics of robots, Jacobians, singular configurations, dynamic and non-differentiable constraints, time-parametrised trajectories, splines and polynomial trajectory generation, minimum jerk solutions, planning for mobile robotic systems. Real-time Environment Interaction. Multimodal sensor fusion, filtering techniques, real-time obstacle detection and collision avoidance, simultaneous localisation and mapping (SLAM), visual odometry, grid and topological maps, occupancy grids. Hands-on Projects. Introduction of common planning and optimisation libraries, cases of study, examples, and end of course group project.

Teaching/Learning Methodology	Lectures will be used to provide students with theoretical knowledge required for an advanced in-depth study of the field of robot motion planning and navigation. (learning outcomes a to d)							
	Assignments and group projects will be used to provide students with a hands-on experience to solve practical problems using the acquired knowledge. (learning outcome a to e)							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	E	
	1. Assignments	30%	✓	✓	✓	✓	✓	
	2. Project	20%	✓	✓	✓	✓	✓	
	3. Class test	10%	✓	✓	✓	✓		
	3. Final exam	40%	✓	✓	✓	✓		
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	Overall Assessment: 0.40 x End of subject exam + 0.60 x Continuous assessment.							
	1. The continuous assessment aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.							
2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes								
Student Study Effort Expected	Class contact:							
	▪ Lecturers						33 Hrs.	
	▪ Tutorials / Exercises / Laboratory						6 Hrs.	
	Other student study effort:							
	▪ Self-learning						36 Hrs.	
	▪ Assignments and project						40 Hrs.	
	Total student study effort						115 Hrs.	

Reading List and References	1. LaValle, S. M. Planning Algorithms. Cambridge University Press., latest edition. 2. Choset, H., et al. Principles of Robot Motion: Theory, Algorithms, and Implementations. MIT Press., latest edition 3. Latombe J.-C., Robot Motion Planning, Springer, latest edition. 4. Siegwart, R., Nourbakhsh, I., & Scaramuzza, D.. Introduction to Autonomous Mobile Robots. MIT Press., latest edition.
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